



Department of Information Technology

Academic Year 2024 – 2025(Odd Semester)

Degree, Semester & Branch: B.Tech-V & IT

Course Code & Title: CCS334 & Big Data Analytics

Name of the Faculty member (s): M.Rethinakumari

Innovative Practice Description

- **Unit / Topic:** Unit II / NoSQL- Aggregate Data Model
- **Course Outcome:** The students can recall all the types in the Aggregate Model and example tools related to Hadoop.
- **Activity Chosen:** Mind Map
- **Justification:**

The decision to use a mind map for teaching the Aggregate Data Model in NoSQL databases is justified by its ability to simplify complex concepts, enhance student engagement, and support innovative teaching methods. This approach promotes comprehensive understanding, critical analysis, and practical application, aligning with modern educational goals and providing a valuable learning experience for students.

- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - **Topic Chosen:** At the end of the unit, the "Aggregate Data Model in NoSQL" topic was selected for the innovative activity.
 - **Initial Announcement:** The topic was announced to the students, and an explanation of the Aggregate Data Model and its significance in NoSQL databases was provided as a recall as shown in Fig 1.
 - **Objective:** The mind map was introduced as a tool to visually organize and represent the Aggregate Data Model, aiming to enhance understanding and interaction.
 - **Mind Map Presentation:** The structure and content of the mind map were explained to the students, highlighting its central node and key branches.
 - **Understanding the Mind Map:** The purpose of the mind map was explained, including how it would be used to facilitate learning. Students were guided on how to interact with and utilize the mind map as shown in Fig 1.
 - **Preparation Time:** Students were given a set amount of time 15 minutes to explore the mind map, understand its components, and prepare questions or feedback related to the Aggregate Data Model as shown in Fig 2 and 3.



• CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO10	PO11	PSO1	PSO2	PSO3
CO2	3	2	2	2	2	3	2	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO10	PO11	PSO1	PSO2	PSO3
	3	2	2	2	2	3	2	3
Justification for correlation	A basic knowledge of engineering fundamentals is necessary to handle big data.	Big data storage format is important in analysis of complex engineering problems especially many business domains.	Choose appropriate big data storage databases and conduct analysis on complex problems based on the big data.	Able to communicate the decision to various analysts by understanding underlying databases.	Able to analyze a huge amount of data and make business decisions and communicate with team members.	Able to analyze huge amount of data to solve complex computation task.	Able to design and analyze the data to develop reliable IT solutions.	Able to determine and apply appropriate techniques for providing solutions to real world problems in Industry and Research establishments.

• Images / Screenshot of the practice:

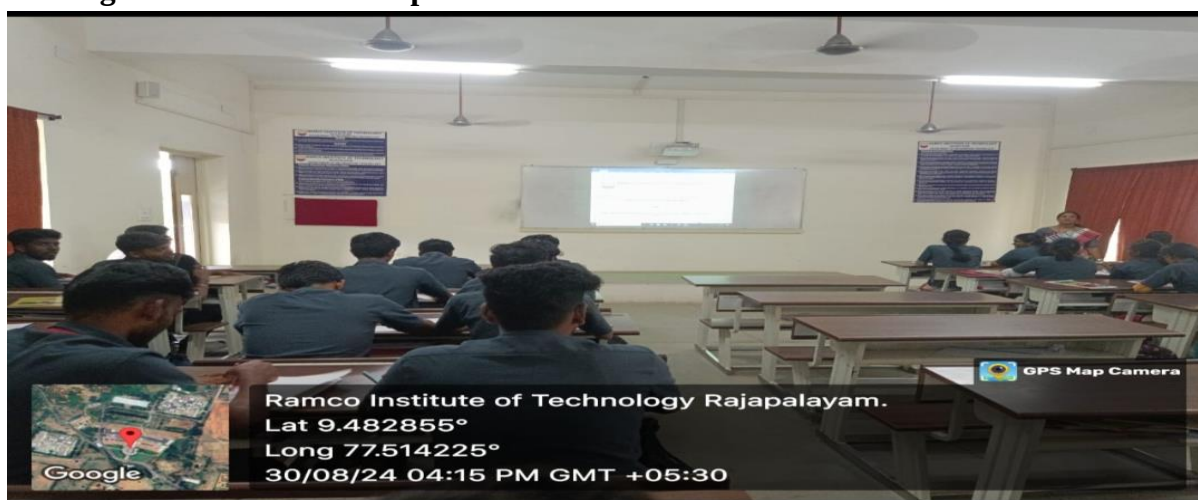


Fig 1: Problem Statement: Recalling the Topic-Aggregate Data Model



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Fig 2: Students working on creating Mind Map for Aggregate Data Model



Fig 3: Collecting feedback and discussing the corrective actions

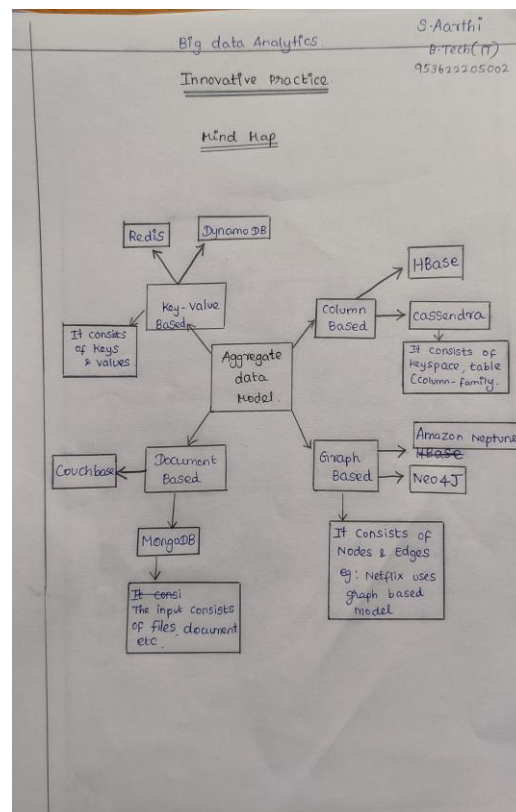
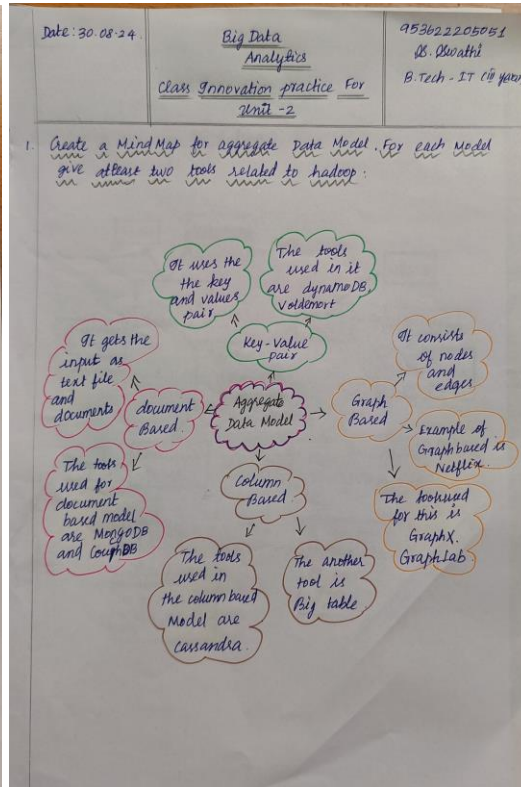
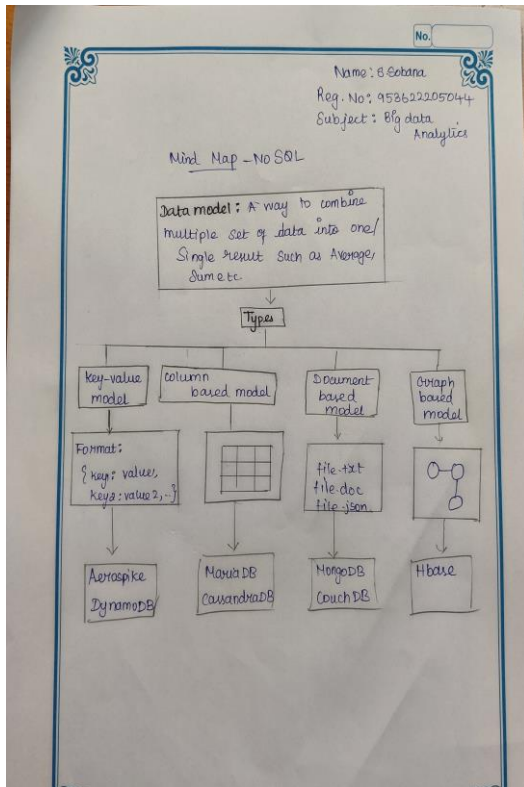


Fig 3: Mind Maps Presented by Students: S. Sobana; S. Swathi; S. Arthi



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- This hands-on approach helps demystify abstract concepts and makes learning more interactive and relevant.
- Many students have expressed that the aggregate data model has enhanced their understanding of data management.
- Students appreciate how aggregate data modeling bridges the gap between theory and practice.
- This practical application helps solidify their grasp of theoretical concepts.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students felt this understanding extends beyond theoretical knowledge to practical application.
- Students find this aspect of aggregate data modeling particularly relevant, as it mirrors the need for adaptability in software development and data management.

- ❖ ***Challenges faced in implementation:***

- Feedback indicates that the students sometimes struggle with understanding how these models affect data accuracy and system performance. Additional resources or practical exercises could enhance their comprehension.
- Some feedback suggests that integrating aggregate data modeling with other relevant topics, such as distributed systems or data consistency, could provide a more holistic understanding. Students feel that connecting these concepts can enhance their ability to see the bigger picture and how different aspects of data management interrelate.

- **References:**

[NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence](#) by Pramod J. Sadalage and Martin Fowler. Addison-Wesley, 2012.

[Innovative Teaching Strategies: A Guide to Enhancing Student Engagement](#) by David W. Johnson and Roger T. Johnson. Routledge, 2015.

Signature of Faculty Member

HOD